MICROPROSE · 1994 · MS-DOS

Sid Meier's COLONIZATION

Technical Reference

The shipped MS-DOS release, documented from its binaries and data files. Executables and overlay architecture · asset containers · terrain and the map compositor · colonies, market, and the native economy · units, movement, and combat · powers, diplomacy, and revolution · events and strings · the UI engine, screen by screen · the map editor · data structures.

RECONSTRUCTED BY DISASSEMBLY OF VICEROY.EXE AND MAPEDIT.EXE · EVERY FIGURE TRACED TO A FILE OFFSET, A DATA-FILE FIELD, OR LIVE GAME MEMORY — OR MARKED UNMAPPED

Contents

The shipped 1994 MS-DOS release, documented from its binaries and data files. Facts in this volume were established by disassembly of the shipped executables, by reads of live game memory under emulation, and by pixel-level comparison of screens reconstructed from these pages against the running game. Where a value could not be established it is marked unmapped or runtime — no figure in this volume is invented.

PART III

Economy and colonies

§9 Market and trade

The European market is per-power state inside the PowerRecord (one 316-byte record per European power; the active power is the pointer `power`). Prices are not a fixed table: the per-good base is random-seeded at game start and then driven by a per-turn decay plus per-transaction updates, with the four manufactured luxuries coupled through a shared supply pool.

9.1 Per-power money and market state

FIELD	TYPE	MEANING
`+0x01`	u8	**tax rate** (0..100) — read for the Europe banner; raised by King events; hard-clamped at 75
`+0x1E`	u16	artillery-bought counter (Europe artillery price escalation: `cost = base + count·100`; incremented on each purchase, zeroed at new game)
`+0x20`	u16	**boycott bitmask**, bit = good index. Test is_boycotted (`(1<<good) & [bx+0x20]`); set (Tea Party); back-tax lift (`&= ~bit` after paying count×500 into the King fund `+0x22`); **Jakob Fugger** (Founding Father id 1) clears the whole word to 0
`+0x22`	s32	King's REF fund (receives sale tax — §18.10)
`+0x26`	s32	sales tally (net proceeds accumulator)
`+0x2A`	u32	**gold (treasury)** — every credit goes through a clamp helper: add s32, clamp to [0, 999999]
`+0x4C`	u8[16]	per-good **price level** — the internal price. Display: sell = level − 1, buy = level + burden
`+0x5C`	s16[16]	per-good **traffic accumulator** ("market pool") — the drift's pressure counter. **Negative → the price is about to rise; positive → about to fall**; steps fire at ±100·(rise|fall)
`+0x7C`	s32[16]	**traded volume** — cumulative *value* (price×qty; sales net of tax). Write-only in the decoded image — its reader is unlocated (TBD)
`+0xBC`	s32[16]	**European supply** per good, in units (sell +qty / buy −qty). Reader likewise unlocated (TBD)
`+0xFC`	s32[16]	per-good **net-trade accumulator** — summed (clamped ≥ 0) by the drift each turn. **Never cleared during play** — a whole-game counter, which is what makes the seed decay a slow secular drift

The four 16-entry arrays are all zeroed at new game; the price levels are then rolled per good — **one roll shared by all four powers** — as `level = start1 + random(0 .. start2 − start1)` from the `@CARGO` table below.

```
sell price = level − 1
buy price = level + burden
visible spread = burden + 1
```

EXAMPLE Food at level 9 with burden 7 → sells at 8, buys at 16 — the spread column below is why Food and Lumber feel wide

(Correction 2026-08-01: earlier printings derived the spread from the wrong `@CARGO` column — the loader stores nine fields per good and the buy-side constant is field 4, "burden", exactly as the NAMES.TXT legend defines it.)

9.2 The sixteen goods — the complete market table

The 16 goods — ICONS.SS frames good+0x17 NAMES @CARGO order · disk 22–37

Good id indexes every market array, the stockpile bar and the boycott bitmask; icons drawn at colony/Europe bars y=181.

Good ids 0..15 in NAMES `@CARGO` order; every market array uses this index. The nine fixed columns come from `@CARGO` itself (parsed at boot — the numbers live in NAMES.TXT, not the EXE); the last five are the live per-power state, shown with sample values†.

GOOD	START	LOW–HIGH	SPREAD	RISES AT	FALLS AT	DRIFT/TURN	SHIFT	PRICE†	POOL†	TRADED†	SUPPLY†	NET TRADE†
Food	1–3	1–6	8	−300	+200	−1	0	1	−40	1,900	600	450
Sugar	4–7	3–7	2	−400	+600	−8	1	7	310	5,200	890	780
Tobacco	3–5	2–5	2	−400	+800	−10	1	5	150	6,000	1,140	1,210
Cotton	2–5	2–5	2	−400	+600	−11	1	5	85	3,100	700	640
Furs	4–6	2–6	2	−400	+2,000	−13	1	6	990	7,400	1,530	1,480
Lumber	2	2–2	5	−300	+200	0	0	2	0	180	90	60
Ore	3–6	2–6	3	−200	+400	−7	0	6	−25	900	260	210
Silver	20	2–20	1	−800	+100	−8	2	20	60	8,100	410	405
Horses	2–3	2–11	1	−300	+200	−3	0	5	−130	−2,300	−380	−460
Rum	11–13	1–20	1	−400	+400	−12	1	13	220	9,800	820	760
Cigars	11–13	1–20	1	−400	+400	−11	1	11	180	4,300	400	390
Cloth	11–13	1–20	1	−400	+400	−13	1	13	140	5,600	430	420
Coats	11–13	1–20	1	−400	+400	−11	1	9	90	2,700	310	300
Trade Goods	2–3	2–12	1	−200	+300	+4	0	2	−60	−1,100	−540	−520
Tools	2	2–9	1	−200	+200	+5	0	2	−190	−2,600	−1,240	−1,290
Muskets	3	2–20	1	−200	+200	+6	0	7	−240	−5,500	−900	−830

Column key — **start**: the new-game roll window `[start1, start2]` (one roll for all powers). **low–high**: the price floor and ceiling the stepping respects. **spread**: burden + 1, the visible bid/ask gap. **rises at / falls at**: the traffic-accumulator thresholds ±100·(rise|fall) — the price steps 1 and the accumulator gives the threshold back. **drift/turn**: the attrition added to the accumulator every turn (negative = drifts toward a rise; Trade Goods/Tools/Muskets drift toward *falls* — the Crown's manufactures cheapen). **shift**: the volatility exponent — each traded unit counts `qty << shift` in the pool. †**sample columns**: *price* is a captured mid-game snapshot (a Dutch game); *pool / traded / supply / net trade* have no captured values anywhere in the repo (all four start at zero and evolve live), so the figures shown are **illustrative samples** on the engine's own scales — positive where a colony typically net-sells, negative where it net-buys — not byte-verified data (samples needed — TBD).

Idle-market cadence falls straight out of the table: with no trade at all, Sugar rises every 400/8 = 50 turns, Furs every ≈31, Silver every 100 — while Tools and Muskets *fall* on the same clock. Four **extended ids 16..19** are name-only production tokens (Hammers, Crosses, Liberty Bells, Flags) — never traded, used for production display. Commodity icons are ICONS frames `good + 0x17` in EXE-sheet numbering.

9.3 Price drift

Per-turn driver: the silent all-goods drift rides check_immigration() inside the production phase — one drift_prices pass for the human power, then the immigration crosses check. (The four-power *interactive* drift loop runs once at game creation, from new_game_power_init — the routine formerly mislabelled "market_day"; see the 2026-08-01 ruling.) **Per-transaction**: the SELL handler (sell_goods) and BUY handler (buy_goods) each call `drift(good, 0)` — a single-good re-drift immediately after the trade.

The drift itself (drift_prices):

```
for good in 0..15:
    acc = price_seed[good]              # word table, indexed good·2
    for power in 0..3:
        v = PowerRecord[power].accum_FC[good]
        if v < 0: v = 0
        acc += v                        # 32-bit
    if driver-mode:
        price_seed[good] -= acc >> 8    # proportional decay
```

`price_seed[16]` is **randomized at new-game init**: seed_market fills each entry with `random_int(600, 1000)` — there is no fixed base-price table. Later phases of drift_prices couple the luxuries: `S_pair = supply[9]+supply[10]+supply[11]+supply[12]` (Rum, Cigars, Cloth, Coats; 32-bit adds, clamp ≥ 1), then for each finished good `target[i] = (S_pair·3)/supply[i]` (×3, 32-bit divide); the raw inputs 1..4 (Sugar/Tobacco/Cotton/Furs) use the same formula against their own supply (Furs halved, +1 if year<1700 and +1 if year<1600). Dumping one luxury lowers its own price and nudges the other three up.

each turn: price_seed -= (price_seed + Σ positive trade accumulators) ÷ 256
luxury coupling: target = (S_pair × 3) ÷ own_supply, S_pair = Rum + Cigars + Cloth + Coats supply

EXAMPLE seed 800 and the four powers sold a net 240 Food → 800 − (800+240)÷256 = 800 − 4 = 796

EXAMPLE supplies Rum 400 / Cigars 300 / Cloth 200 / Coats 100 → S_pair 1000 → Rum's target = 3000÷400 = 7

The stepping rule (per good, per power): each turn the accumulator gains the drift/turn column; each trade adds `±(qty << shift)` plus a value term. When the accumulator reaches **−100·rise** the price steps **up** 1 (max = high) and the threshold is credited back; at **+100·fall** it steps **down** 1 (min = low). Each step fires `@PRICEUP`/`@PRICEDOWN` for a human power — so the message cadence in a quiet market is exactly threshold ÷ |drift|.

Per-good specials, all byte-traced: **Tools/Muskets** get a time-and-difficulty ceiling bump `(difficulty − 4)·2 + (year − 600)/100`, and Horses/Tools/Muskets carry a hard published ceiling `max(level, (−(difficulty − 4)·3)/2 + 3)`. **Furs** supply is halved in the luxury-coupling pass, +1 before 1700 and +1 more before 1600 — the early fur boom. **Food** is excluded from the raw-goods repricing loop entirely; its price moves only through the accumulator path. **Lumber** has zero drift — its price never moves on its own. **Silver** starts pinned at its ceiling of 20. The **Dutch** trade twice as calmly: their sell-pool delta is ×2/3, and their attrition is **doubled on odd turns** — prices recover toward the Dutch twice as fast (previously undocumented).

9.4 Buy/sell transactions

SELL (sell_goods; args good, screen-idx, confirm): `gross = price·qty`; tax split: `tax = gross·tax_rate/100`, `net = gross − tax`; the treasury is credited `+net` through the [0,999999] clamp helper; the King's fund gains the tax; the sales tally gains the net. **BUY** (six inline sites in the purchase pages, e.g. Muskets qty 50, Horses, Tools qty 100): affordability check then an inline 32-bit debit of the treasury — **buys are untaxed**.

tax = gross × tax_rate ÷ 100

you receive gross − tax the King's fund gains the tax

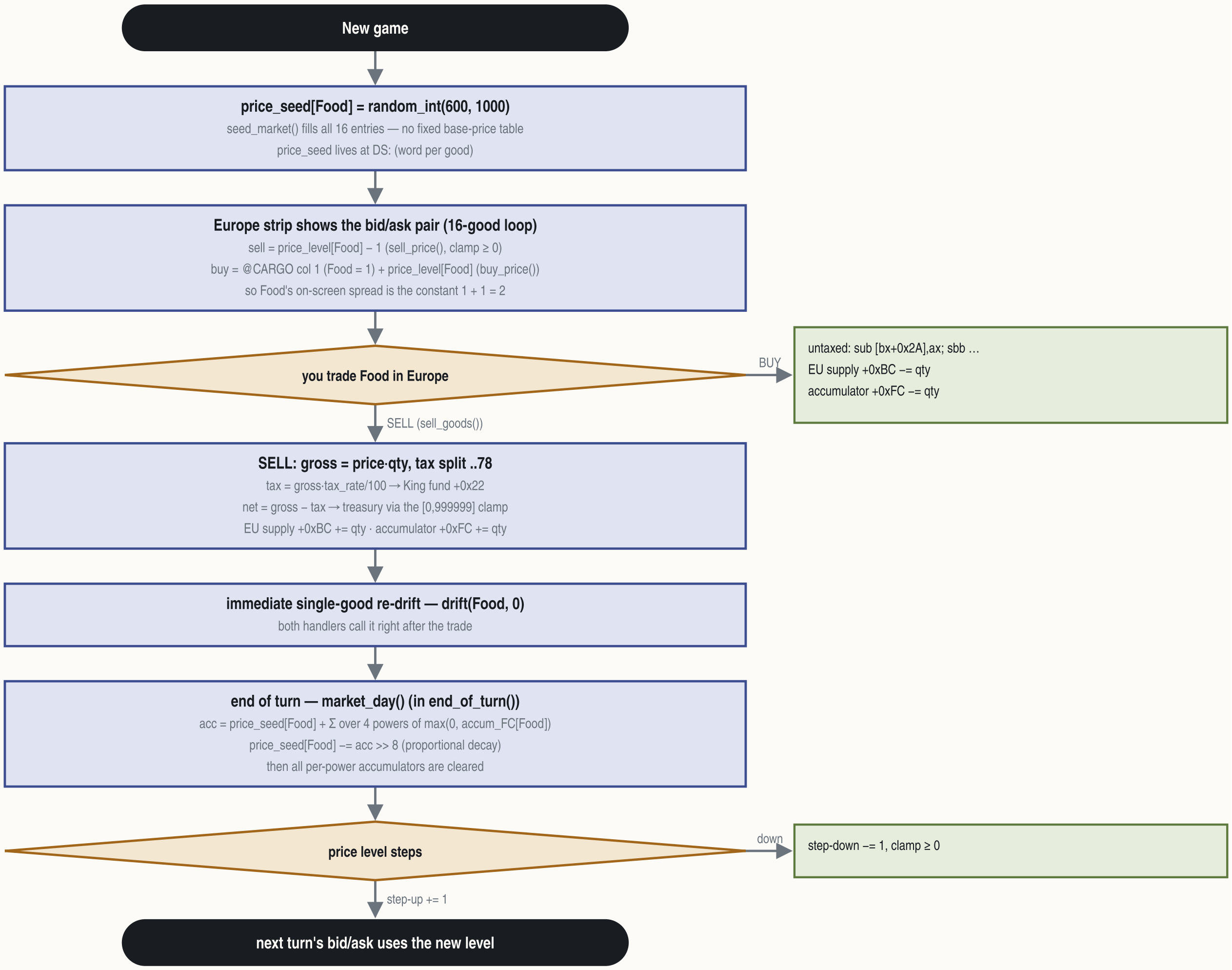
EXAMPLE sell 100 Muskets at 12 gold with tax at 33% → gross 1,200 → 396 to the King, 804 to your treasury

Both paths call a mirror pair of accumulator updaters (good, qty):

FIELD	BUY RECORD_PURCHASE	SELL RECORD_SALE
EU supply `+0xBC`	`-= qty`	`+= qty`
accumulator `+0xFC`	`-= qty`	`+= qty`
traded volume `+0x7C`	`-= price·qty`	`+= price·qty·(100−tax%)/100`

Walkthrough — the life of the Food price

good 0 · assembled from §9.1 / §9.3 / §9.4



Food is not luxury-coupled. Rum/Cigars/Cloth/Coats additionally share S_pair = supply[9]+…+supply[12] with target[i] = (S_pair·3)/supply[i] ; raw inputs Sugar/Tobacco/Cotton/Furs use the sameformula against their own supply (Furs halved; +1 if year<1700, +1 more if year<1600,).

One asymmetry worth knowing: the colony warehouse **overflow auto-sale** (goods over 100 cut to 50, the excess sold and taxed — §18.10) credits gold and the King's fund but does **not** run these accumulator updaters — forced exports never move the market. | DGROUP pool (4 records, stride `0x9E`) | `-= scaled_qty` ×4 | `+= scaled_qty` ×4 (4th power ×2/3) |

`scaled_qty = ((price_level−2)·16·qty)/100`; the `@CARGO` "spread" column (field 4) is a per-good left-shift exponent on qty inside these updaters, not the display spread.

On the Europe screen the market bar (0,179,320,21; 16 cells, pitch 19, icons `good+0x17`, bid price centred at y=194) routes clicks to the sell handler, which first tests the boycott bit and blocks with a message if set. Recruit gold cost comes from the recruit-pool slot word (slot stride 6, cost in the word at offset 4) — for Artillery (colonist type 11) it escalates `base + artillery_bought·100`.

PART V

Politics and powers

§18 Revolution, the King, and multiplayer

The endgame pivots on one meter, one flag word and one power id: the national Sons-of-Liberty meter `game.revolution_meter` (0..100), the game-state bits `game.flags`, and the King/REF power `game.king_power`. Declaring independence flips `game.flags` bit 0, turns the Crown's standing Expeditionary Force into an on-map power, and is won by attrition, not by timer.

18.1 The revolution meter game.revolution_meter

Initialized to 0 at new game; Bolívar adds +20 capped at 100; the king's tax-severity score reads it. The endgame dispatcher clamps it to 75 and routes: below 75 with `game.king_power` < 0 → the Spanish-Succession arm; at/above 75 → the revolution handlers. In hot-seat multiplayer the pre-war state is held at this 75 cap and the auto-revolution arms are suppressed.

18.2 Declaring independence

EVENT_ID	STRING_KEY	TRIGGER	CONDITION	OPTIONS	OUTCOMES	ARMS
declare-1	`@ALREADYREVOLUTION`	Declare-Independence command → declaration_gate	`game.flags` bit 0 already set	—	none (return)	—
declare-2	`@TOOTORY`	same	`game.revolution_meter` < 50	—	shows "Only {%NUMBER0%%} of the colonists support the independence movement… We cannot start a rebellion against the King until the {majority} is behind us."; return	—
declare-3	`@MULTIREV`	same, hot-seat only	hot-seat flag set	"Declare independence" / "Never Mind"	on confirm the hot-seat flag is cleared — the game demotes to single-player exactly as the text warns	falls through to declare-4
declare-4	`@DECLARE`	same	SoL ≥ 50	"Never! That would be treasonous! God save the King!" / "Yes! Give me liberty or give me death!"	rebel power `game.rebel_power` := `game.current_power`; on row 2 → declare_independence	declaration
declare-5	`@INDEPENDENCE`	declare_independence	—	—	`game.flags` |= 1; the declaration year is recorded; initial REF dispatch	war

18.3 Game-state bits game.flags (word; save block #3)

BIT	MEANING
0	War of Independence declared (stage 1)
1	foreign intervention active (stage 2; set by declare_intervention)
3	independence WON (gates the score bonus)

BIT	MEANING
TBD	REF-arrival phase flag; also gates the Congress driver off
5	forced/end-stage flag (set by the dispatcher once SoL≥75+declared+intervention, and by `@FORCED` stage d)
high bits	the Game Options word (high-byte rows: Tutorial Hints, Water Cycling (inverted), Combat Analysis, Autosave, End of Turn, Fast Slide, cheat master, Show Foreign, Show Indian)

Victory: the per-turn resolver (runs while bit 0 set, bit 3 clear) counts surviving King-owned units of types {6, 8, 0xB}; when the count falls below the threshold (1 normally, 8 when a specific flag bit is set) and the intervention tally clears, the independence-won bit of `game.flags` is set — the rebels win, with the message built from the rebel power's personality record. If independence is won and was declared before 1780, the score gains `(1780 - declaration_year) * 2`.

18.4 The King's power and the @FORCED staged advancer

The REF exists pre-war only as the four count globals, seeded at new game by new_game_setup (`8d+15 / 5d+5 / 3d+2 / 6d+2` for difficulty d) and grown by grow_royal_fund (the royal fund accrues `(8*diff+10)*2^era` per turn; at ≥ 1800 one unit is bought, 1800 subtracted, and the slot chosen by ratio). At the war transition the Crown becomes a real on-map power whose id lands in `game.king_power`.

Cheat id 0x68 "Advance Revolution Status" (DEBUG.TXT `@FORCED`) advances one stage per invocation and documents the staging exactly:

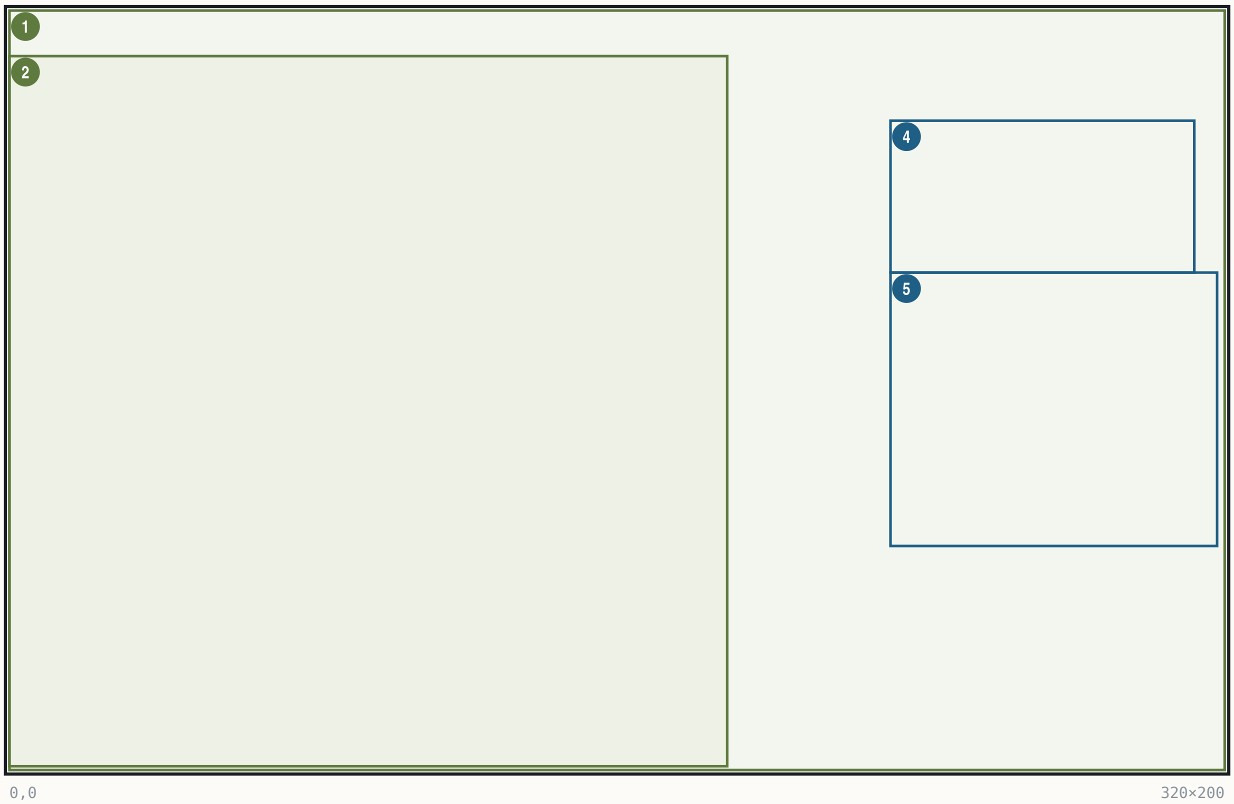
STAGE	CONDITION	ACTION
a	`game.revolution_meter` < 75	set the meter to 75 and create the REF power if none (via spanish_succession)
b	—	declare independence (declare_independence; `game.flags` |= 1)
c	—	next war stage (declare_intervention; `game.flags` |= 2)
d	—	`game.flags` |= 0x20 and show the `@FORCED` text

Stages b–d are blocked in hot-seat multiplayer.

18.5 The King audience screen

One renderer paints the audience/tax-demand, the loss and the win screens. Assets: backdrop **KINGLSS1.PIK** (throne room, empty chair, blank scroll); outcome-selected foreground sheet **KING1.SS** (audience) / KINGLOSE / KINGWIN — the king-and-dog figure, 189×187 — plus the nation banner sheet (nation stem + digit, e.g. ENGLND1.SS, the throne-canopy banner). The variant is selected from the two stack args; the nation prefix comes from `game.rebel_power` (ENGLND/FRANCE/SPAIN/DUTCH). Callers: the audience sequencer (variant 1,1 → KING1) and the King-event orchestrator periodic_phase (loss, win).

The King audience screen 320×200 · Mode 13h



#	BOUNDS	KIND	REGION	NOTE
1	(0,0) 320×200	art	KINGLSS1.PIK throne room	full-screen backdrop
2	(0,12) 189×187	art	KING1/KINGLOSE/KINGWIN.SS king	bottom-anchored to row 199 (frame-descriptor anchor)
3	(32,0) ·×· ᴿ	art	ENGLND<d>.SS canopy banner	nation-selected; size from sheet
4	(232,29) 80×40	text	speech header, 4 lines	per-line centered on x=271.5, tops y=29..61 (measured; not byte-cited)
5	(232,69) 86×72	text	speech body, 9 lines	left x=232, pitch 8 = FONTKING H+1 (measured; not byte-cited)

FONTKING is loaded; the stored pen values (242,47) are engine register values, not the on-screen origin — the glyph runner relays the text under its mode flags. Body text is runtime-built by periodic_phase from the GAME.TXT tax family; the pen/ink is restored to the FONTINTR color on exit. Fade-in via the palette verb.

Tax-event branch keys (GAME.TXT, `@width=190`, speaker channel 8 → KING1.SS): `@KINGTAX` ("…we have graciously decided to raise your tax rate by {%NUMBER0%%}. The tax rate is now {%NUMBER1%%}. If you wish, you may kiss our royal pinky ring."), `@KINGRAISE` (punitive raise for demanding lower taxes), `@KINGLOWER`, `@KINGNOTHING` ("…You may, however, kiss our royal pinky ring."), `@MERCANTILISM`, `@PURCHASETAX`, pretexts `@KINGWIFE`/`@KINGWAR`/`@KINGNAVACT`/ `@KINGSTAMPACT` (severity-selected, section on taxation), options `@TAXOPTIONS` ("Kiss pinky ring." / "Hold '{%STRING3 Party}.'") and `@TEAPARTY`.

18.6 Hot-seat multiplayer

- **Unlock:** environment `SET COLONIZE=MULTI` (checked via `getenv`) sets the multiplayer-unlock flag (a CLI switch exists as well), adding a 5th game-start menu entry (mode 4).
- **`@MULTI`** (new-game setup): "Select powers to be controlled by human players." — a checkbox dialog; each checked power gets controller byte 0 (read from the checkbox bitmask); more than one human sets the hot-seat flag; none checked defaults to England.
- **`@MULTINEXT`** (turn loop): between human turns the screen blanks and shows "^ ^ {%STRING0} Player Turn … Press any key for {%STRING0} player's turn." (advisor 2), then the view power switches and re-centers.

- Consumers of the hot-seat flag: rebel sentiment clamped to 75 with auto-revolution suppressed; `@FORCED` stages b–d blocked; the Spanish-Succession merge skipped; `@MULTIREV` demotes the game to single-player on a confirmed declaration (18.2).

18.7 The War of the Spanish Succession merge

How it triggers (byte-traced 2026-08-01 — this reverses an earlier "unresolved residual" verdict; see the ruling). The merge has three ways in:

1. **The real per-turn trigger — update_sol().** Every turn, inside the production phase, the engine recomputes each power's Sons-of-Liberty percentage. For the human (rebel) power it also copies the result into the national revolution meter — and then fires the succession when **both** hold: **SoL ≥ 50 %**, and **no power has withdrawn yet** (`game.king_power` still negative). There is no dice roll and no year or turn test — crossing 50 % rebel sentiment while all four powers still stand is the entire condition. It can only ever happen once: the handler stores the ceding power's id into `game.king_power`, which permanently fails the second gate.
2. **Forced at the Declaration.** Declaring independence while no power has withdrawn runs the merge immediately — this is how the withdrawn/King's power comes to exist for the war.
3. **The cheat** — menu id `0x68` `@FORCED` stage (a) (§18.4). The "meter clamped to 75" behaviour belongs to *this* staged advancer, not to the gameplay trigger (the source of the old mis-attribution).

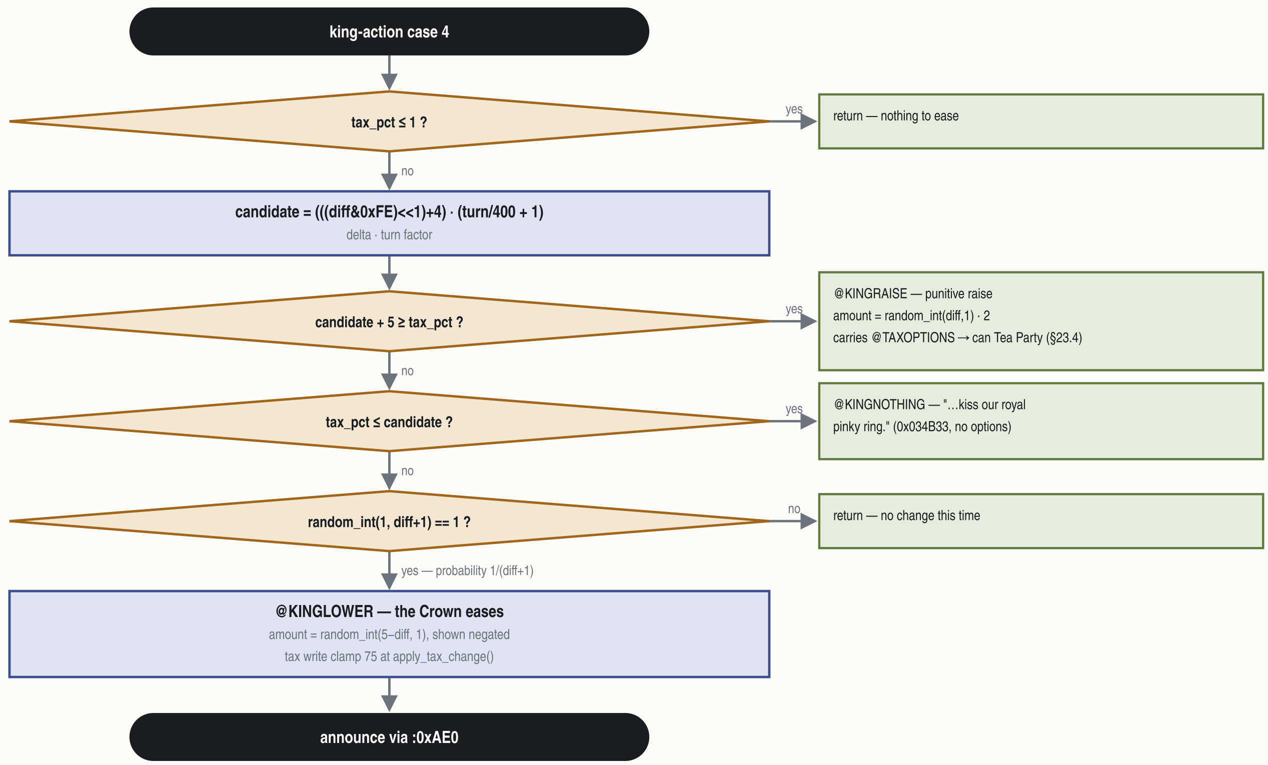
The handler itself runs in single-player only (an entry test).

> **`succession fires when SoL% ≥ 50 and no power has withdrawn yet`**
> `checked every turn as the meter updates; once ever; single-player only`
>
> **EXAMPLE** your rebel sentiment reaches 50 % in 1595 with all four powers alive — the weakest AI cedes its New World to the strongest, that same turn

What it does. It ranks the four powers by a weighted score (3 × one per-power table + 2 × colony count + 1 × a second table), picks the **weakest eligible AI** as the ceding power and the **strongest eligible AI** as the beneficiary, emits `@SUCCESSION` ("War of the Spanish Succession ends in Europe! {%STRING0}, ravaged by war, agrees to cede %STRING1 to the {%STRING2}. Treaty of Utrecht specifies that all {%STRING3} possessions in the New World now fall under {%STRING2} rule."), then rewrites every map-tile owner nibble, every unit owner, every colony owner, and a third owner-nibble table. Aftermath: the ceding power's controller becomes "eliminated", its id lands in `game.king_power`, and it thereafter renders as "(Withdrawn from New World)". What transfers is exactly tiles, units, colonies and the third owner table — **no treasury, no Founding Fathers, no trade routes, no Europe dock**.

18.8 The tax petition — how the King eases (or punishes)

The tax petition — three outcomes tax_petition()



No player-facing 'request lower taxes' control is documented; the announce overlay's internal tax write is untraced (TBD).

Handler tax_petition (dispatched as case 4 of the king-action switch next_immigrant_class, inside turn phase 4). Three guards, then a roll, then one of three outcomes:

- **Guard 1**: return if `tax_pct ≤ 1`. **Candidate** raise = `delta·turn_factor` with `delta = ((diff & 0xFE) << 1) + 4` and `turn_factor = turn/400 + 1`.
- **Guard 2**: if `candidate + 5 ≥ tax_pct` → **@KINGRAISE** path.
- **Guard 3**: if `tax_pct ≤ candidate` → **@KINGNOTHING** ("…kiss our royal pinky ring.", no options).
- **Roll**: `random_int(1, diff+1)`; result 1 → **@KINGLOWER** (probability `1/(diff+1)` — the Crown eases more readily at low difficulty). Lower amount = `random_int(5 − diff, 1)` negated for display ("…lower your tax rate by {%NUMBER0%%}").
- **@KINGRAISE** amount = `random_int(diff, 1) · 2` — the punitive raise is doubled ("Your DARE to demand lower taxes!…"), and it carries the `@TAXOPTIONS` pair, so it can chain into a Tea Party (§23.4).

Open items, stated plainly: no player-facing "request lower taxes" control is documented anywhere in the UI specs (the "player demands" framing is narrative); the outbound announcement goes through an overlay whose internal tax write is untraced; the per-turn call frequency of tax_petition is unmapped. The tax write-site clamp itself is byte-cited: any applied delta lands at apply_tax_change with the hard clamp at 75.

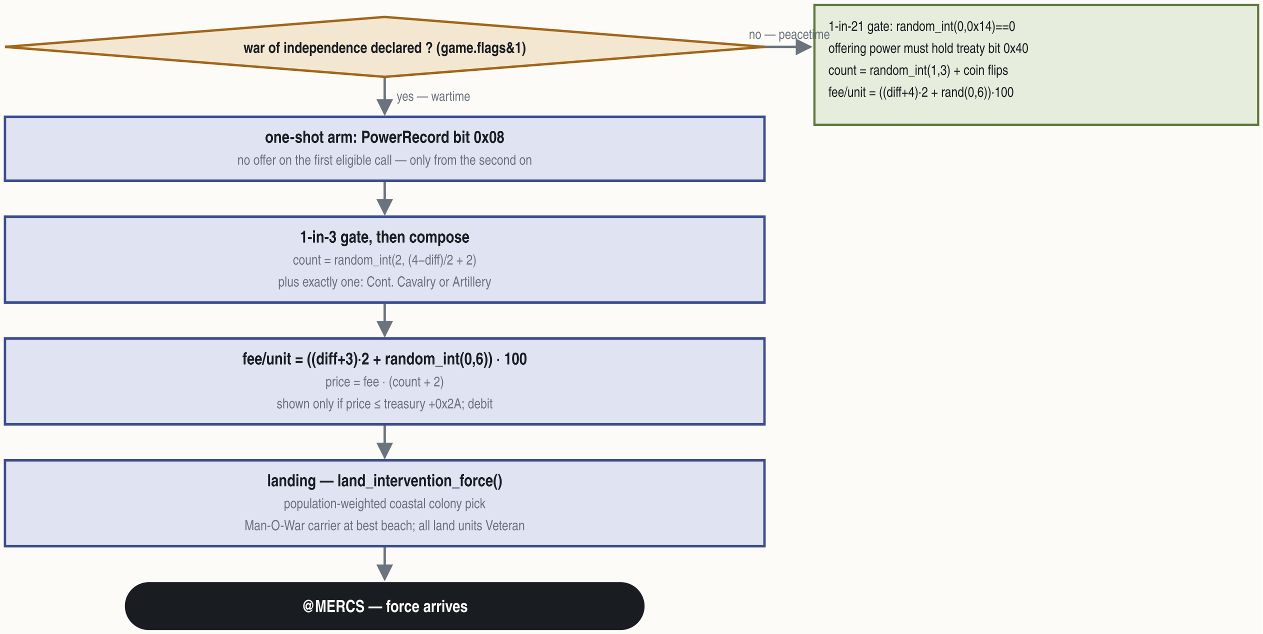
```
candidate raise = ( ((d & 0xFE) × 2) + 4 ) × ( turn÷400 + 1 )
ease amount = random( 1 .. 5−d )
P( ease ) = 1 ÷ (d + 1)
punitive raise = random( 1 .. d ) × 2
```

EXAMPLE Conquistador at turn 500 → candidate (2×2+4)×2 = 16; ease odds 1-in-3, easing 1–3 points off the rate

18.9 The King's mercenaries

The mercenary offer

peacetime king_phase() · wartime offer_wartime_mercenaries()



Two distinct offers, both priced per unit in hundreds of gold. (The manual's §23.4 event table has no mercenary row — this subsection is the coverage.)

Peacetime (king_phase, turn phase 1): gated off once the revolution starts (the war-declared bit of `game.flags`), then a **1-in-21** roll (`random_int(0,20) != 0` → return); the offering power must hold a peace treaty (the treaty bit). Composition: `count = random_int(1,3)` plus coin-flips that either add one more or set 1–2 artillery. **Fee**: `gold_per_unit = ((diff + 4)·2 + random_int(0,6)) · 100`, `price = gold_per_unit · (count + 2·artillery)`. Offer dialog `@MERCENARIES` ("The King of %STRING0 has offered to send us a force of trained {mercenaries}… No thank you. / Pay {%NUMBER0$}."); arrival `@MERCS`. Delivered units are Veteran Dragoons and Veteran Artillery on a Man-O-War.

```
peacetime fee = ( (d + 4) × 2 + random(0..6) ) × 100 per unit
```

EXAMPLE Conquistador, middle roll 3 → (12+3)×100 = 1,500 gold per Dragoon offered

Wartime (offer_wartime_mercenaries): a per-power one-shot bit (PowerRecord byte 0, bit `0x08` — set on the first eligible call; an offer is possible only from the second call on), then a **1-in-3** gate. Composition: `count = random_int(2, (4−diff)/2 + 2)` plus exactly one of Continental Cavalry or Artillery. **Fee**: `gold_per_unit = ((diff + 3)·2 + random_int(0,6)) · 100`, `price = gold_per_unit · (count + 2)`. The offer only appears if affordable (price ≤ the treasury); paying debits it. Wartime types: Veteran Continental Army + Continental Cavalry / Artillery.

```
wartime fee = ( (d + 3) × 2 + random(0..6) ) × 100 per unit × ( count + 2 )
```

EXAMPLE Conquistador war offer, roll 3 → (10+3)×100 = 1,300 per unit; 3 troops +2 → 6,500 gold total

Landing (land_intervention_force, shared with the free intervention force): arrival colony is a population-weighted random pick over up to 10 coastal colonies (selected on a colony coastal flag; weights = size), a Man-O-War (type `0x12`) lands at the best-scored beach tile, troops spawn carried at the (−2,−2) sentinel, and every land unit is stamped Veteran.

Mobilization at the Declaration (mobilize_continentals, byte-verified): for every colony with SoL ≥ 50, a budget `((SoL−50)·(size/2))/50`, clamped ≥ 1, of Veteran Soldiers/Dragoons in the stack are promoted in place — type 1 → 9 Continental Army, type 4 → 7 Continental Cavalry; `@MOBILIZE`/`@MOBILIZE2`. No units are created.

18.10 The King's economy — sales tax, the royal fund, and the REF

The Crown runs a closed loop: **your taxes fund the army that will one day be sent against you.** Every gold piece of tax you pay lands in the royal fund (`power.royal_fund`), and every 1,800 banked buys one more Royal Expeditionary Force unit.

Income — the three tax feeds (all byte-traced; the fund is credited at three distinct sites):

FEED	AMOUNT INTO THE FUND
every European sale	`gross × tax% / 100` (you keep the net)
colony warehouse overflow auto-sale	the tax share of the forced sale — goods over 100 are cut to 50 and the excess sold on your behalf
paying back taxes to lift a boycott	the whole payment: `boycotted count × 500`

On top of the taxes the Crown appropriates a **stipend every turn**:

`fund += ( 8·difficulty + 10 ) per turn, doubled at each of 1600 / 1700 / 1750`

`pre-independence only — the fund stops growing the day you declare`

EXAMPLE Explorer pays the Crown 18 a turn in 1550, 36 after 1600, 72 after 1700 — one new REF unit roughly every 25–100 turns before your taxes even help

Spending — one unit per 1,800. When the fund reaches **1,800** it buys exactly one unit that turn (1,800 is both the gate and the price, and it is deducted). Which arm gets it keeps the army in ratio — the checks run in sequence and **each later match overrides the earlier**, so the effective precedence is Man-O-War over Artillery over Cavalry over Regulars:

`buy Cavalry when (regulars + 2) / 3 > cavalry · Artillery when regulars / 4 > artillery · Man-O-War when (reg + cav + art + 5) / 10 > ships · else Regulars`

`the tests run in that order and the LAST match wins — a fleet shortage always outranks a gun shortage, which outranks a horse shortage`

EXAMPLE 24 regulars, 8 cavalry, 5 artillery, 3 ships: cavalry test 8<8 no · artillery 6>5 yes · ships 4>3 yes → the King lays down a Man-O-War

Each purchase is announced with `@KINGBUY` — "*King increases military spending. {unit} added to royal expeditionary force. Colonial leaders express alarm.*" (The `@KINGMOBILIZE` "Parliament votes additional funds" text belongs to the other, wartime arm — the two were swapped in an earlier spec gloss, corrected 2026-08-01.)

The force itself. Counts live in `ref.regulars`, `ref.cavalry`, `ref.man_o_war`, `ref.artillery` (user-verified against the F3 display). The new-game seed by difficulty d:

D	DIFFICULTY	REGULARS 8D+15	CAVALRY 5(D+1)	ARTILLERY 6D+2	MAN-O-WAR 3D+2
0	Discoverer	15	5	2	2
1	Explorer	23	10	8	5
2	Conquistador	31	15	14	8
3	Governor	39	20	20	11
4	Viceroy	47	25	26	14

Three quirks worth knowing: the engine guarantees **at least one Man-O-War** (a zero count is bumped to 1 when the force is totalled); your **own Man-O-Wars are tallied into the King's ship count** by the unit walker; and at the war, units leave the pool **one at a time as they land** — the count decrements per landing, population-weighted onto your coastal colonies, every land unit arriving as a Veteran. There is no "wave" table — deployment is a per-turn eligibility check with budget `(8 − difficulty) × 10` (the landing spawn coordinates are still untraced — TBD). **You win the war when the King's surviving land force (regulars + cavalry + artillery) drops below 4.**

18.11 The difficulty ledger — every documented d-scaled constant

Difficulty `d` = `game.difficulty` (0 Discoverer .. 4 Viceroy; default 2, written only by the difficulty picker). The term is almost always a **human handicap** — most sites gate on the controller byte and use a fixed constant for AI powers.

Starting conditions: gold 1000 (d=0) / 300 (d=1) / 0 (d ≥ 2) — human only; units = Caravel + Pioneers + Soldiers aboard (Dutch ship → Merchantman), **doubled at d ≤ 1** by a second placement pass; REF seed §18.10; native alarm seed `random_int(0,14) + 2d` for the human; year 1492, map 58×72, price seeds `random_int(600,1000)` ×16. Initial tax rate: no initializer is byte-cited (UI shows 0%) — TBD.

EXAMPLE Explorer (d=1) → 300 gold · combat +3 · treaty 10 turns · REF 18/turn · native training 70% · first-father base 64
EXAMPLE Viceroy (d=4) → 0 gold · combat +0 · treaty 4 turns · REF 42/turn · native training 10% · first-father base 112

MECHANIC	FORMULA IN D
Combat: human handicap	strength += (4−d), both sides
Combat: scaling	strength·d/20
Combat: generic base	d+5
Treaty-respect seed	2·(6−d), halved w/ Franklin
AI war grace period	no AI war before turn 10·(10−d)
AI demand value	·10·(d+8)/100, then +500·(d+1)
AI action gate	roll(1,1000) < 200d+100 → 10..90%
Withdrawal price	25·(d+2)·forces, min 100, ×2 at war
King tax-raise delta	((d&`0x`FE)<<1)+4, ×(turn/400+1)
King demand cadence	interval 18→15/12/9 by era, −(d−2) human
Tax-lower odds	1/(d+1); lower amt random(5−d,1)
Mercenary fee (war)	((d+3)·2 + rand(0,6))·100 per unit
Mercenary fee (peace)	((d+4)·2 + rand(0,6))·100 per unit
Mercenary count (war)	random(2, (4−d)/2 + 2)
REF fund accrual	(8d+10)/turn, ×2 per era
REF seed	8d+15 / 5(d+1) / 3d+2 / 6d+2
Native attitude (human)	2·(d+3) + tribe terms, thr `0x`41
Native attitude (AI)	tribe terms − d + 12, thr `0x`32
Native training success	roll(1,1000) ≥ 200d+100 → 90..10%
Native attack chance	random((5−d)·2)
Raze gold factor	rolls of random(1, 10−d)
King treasure cut	max(5d+50, 2·tax) ≤ 90%, tax w/ Cortés
Native gift/reward	2d+15; 10·(d+rand); cap 8−d
Immigration threshold	·(8−d)/8 (England then ×2/3)
Rival-immigration bonus	100·(d+1)
FF bell cost (human)	(d+3)·16 base (§17)
FF cost post-declaration	d·1500 + 2000
FF score penalty	ff_count·(−1−d)
SoL production divisor	10−d (human; AI fixed 10)
Tory penalty threshold	10−d as count threshold
Tory uprising gate	fires with prob (d+1)/(d+2)
Tory militia strength	pop·tory%·2/100 + d + 1
Center-tile food	+2 at d=0, +1 at d=1
Score multiplier	[4,5,6,8,10] = d+4+(d≥3)+(d≥4)
Indian razes score penalty	razed_count·−(1+d)
Tutorial build warnings	only at d < 2

(The Tory militia strength formula is recorded in the rulings batch rather than at a single site.)

Non-mechanics, for the record: a per-difficulty table holds the king salutation/title pointer (text only); Lost-City-Rumor odds are scout-scaled, **not** difficulty-scaled; European recruit prices come from a pre-filled pool word (only artillery escalates, +100·bought).

18.12 The Tory uprising

Processor tory_uprising (its caller resolves through runtime dispatch — no static call site exists). There is **no SoL threshold gate** on the trigger path (byte-verified negative); the only gate is the per-call roll `random_int(0, d+1) != 0` — fire probability `(d+1)/(d+2)`, 50% at Discoverer up to ~83% at Viceroy. Target = the rebel colony with the highest **tory strength** = `pop·(100−SoL%)·2/100 + d + 1`, skipping colonies already hit (a colony flag bit set on the winner). Militia spawn on the free adjacent tiles: `spawn_unit(type 1 Soldiers, owner = the King's power` game.king_power `)`, with a random upgrade gate promoting a spawn to Dragoons; if no adjacent tile is free the uprising is silently suppressed. Message `@TORYUPRISING` ("Tory uprising near %STRING0! Parliament arms Tory Militia!"). The in-repo wording conflict on militia count (≤ 8 per free tile vs strength-counted-down) is unresolved — flagged.

```
tory strength = pop × (100 − SoL%) × 2 ÷ 100 + d + 1
P( fires ) = (d+1) ÷ (d+2)
```

EXAMPLE pop 10 at 30% SoL, Governor (d=3) → 10×70×2÷100 = 14 → +4 = 18 militia strength; fires 4 turns in 5

18.13 Scoring and the Hall of Fame

The final score func_039EE2 components · func_03A9C0 scaler

```
score = (mult · Σ7 terms) / 100 >> 1
mult = d+4 (+1 if d≥3) (+1 if d≥4) = 4 / 5 / 6 / 8 / 10
```

TERM	MEANING	SITE
`population`	+1 criminal/servant/convert · +2 Free Colonist · +4 specialist	`0x3A0BE..0x3A113`
`fathers`	+5 per Founding Father	`0x03A2BE`
`sentiment`	+ national SoL meter [`0x53D0`]	`0x03A561`
`razes`	razed-settlement count × −(1+d)	`0x03A4B1..0x03A4C5`
`gold`	+ treasury / 1000	`0x3A3F2`
`bells`	+ bells/100, only after foreign intervention	`0x3A704..0x3A724`
`revolution`	+ (1780 − declaration year)·2 if independence won	`0x3A5F5..0x3A609`
`rank`	largest n with n²/3 < score, cap 23	`0x3AA41..0x3AA79`

The retail manual's ×2.0/×1.5 revolution multiplier is byte-refuted — the bonus is additive.

Component sum score_components adds seven terms into the grand total: **population** (+1 per criminal/servant/convert, +2 per Free Colonist, +4 per specialist); **Founding Fathers** +5 each; **rebel sentiment** = the national meter `game.revolution_meter` ×1; **razes** = razed-settlement count (`power.razed_count`) × −(1+d) (one spec sheet glosses this same site as an FF penalty; flagged); **gold** = treasury/1000; **post-intervention bells** = the bell pool/100, gated on the intervention flag; **revolution bonus** = `(1780 − declaration_year)·2`, additive, only if independence was won and declared before 1780 (the retail manual's "×2.0 multiplier" framing is byte-refuted). Scaler compute_score: multiplier `d+4 (+1 if d≥3, +1 if d≥4)` = 4/5/6/8/10, `score = (mult·base)/100 >> 1`, Hall-of-Fame rank = largest n with `n²/3 < score`, capped 23. The Hall of Fame renders on WOODPAN2/WOODPANL in FONTINTR (title gold `0xFC`), persists `HALLFAME.DAT` — 5 shown of 6 records, record stride 42 with the score word at offset 38, descending insertion (hall_of_fame).

```
score = ( multiplier × Σ seven terms ) ÷ 100 ÷ 2
multiplier by difficulty: 4 / 5 / 6 / 8 / 10
```

EXAMPLE 900 raw points at Conquistador → $(6\times900)\div100 = 54$ → $\div2 = 27$; rank = largest n with $n^2\div3 < 27$ → rank 8